

MD SHAKHRUL IMAN SIAM

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RESEARCH INTEREST

Multi-modal Large language Models, Audio and Speech signal processing, Egocentric Video processing, Generative AI for Healthcare.

WORK EXPERIENCE

Research Intern

USA

- Microsoft

June 2025 - August 2025

Research Focus: ASR Privacy, Multimodal LLM, Neural Audio Codec

- **DECAF: Disentangling Encoder for Compressed Audio Features**

- Design a Neural Audio Codec that disentangles speaker identity from voice data and transmits with a very low bitrate.
- Achieved 95% improvement of Equal Error Rate compared to state-of-the-art Neural Audio Codecs with only 5% degradation of relative WER compared to original audio.
- Fine-tune a LLM-based Automatic Speech Recognition System that uses the transmitted codes directly as input without the need to reconstruct to the audio waveform.

Graduate Research Associate, The Ohio State University

Columbus, Ohio, USA

- OSU AIoT and Machine Learning Systems Lab 📄

August 2023 - Present

Advisor: Prof. Mi Zhang

- **A large-scale multi-modal reading dataset of 100 hours of egocentric videos**

Accepted to NeurIPS 2025

- A first-of-its-kind dataset that enables egocentric and spatial AI on the next-generation Meta AI glasses.
- Develop a transformer-based model that detects whether and when a user is reading using different input modalities (e.g., egocentric video, eye gaze, IMU)
- Developing a benchmark using both gaze information and RGB image data to classify reading and non-reading activities.

- **LLM-based Medical Question-Answering Assistant using Retrieval-Augmented Generation**

- Developed a Question-Answering Assistant capable of Medical Domain QA tasks.
- Integrate various RAG techniques like Naive RAG, Graph RAG, and Light RAG.
- Designed a knowledge graph and implemented Vector Database using FAISS index and Pinecone.
- Implement various pre-retrieval optimizations for efficient similarity searching.

- **Multimodal Foundation Model for Respiratory Health Diagnosis**

- Developed a Multimodal foundation model that will capture sounds from a smartphone microphone and perform several downstream tasks like disease (Covid-19, TB, COPD) detection, generating health reports, and Question-answering.
- Utilize LLaMA-3 with PEFT and LoRA techniques to fine-tune multimodal models from cough sounds and self-reported patient text metadata.

Machine Learning Engineer

Dhaka, Bangladesh

- ACI Limited (Full-time)

October 2021 - July 2023

Domain Experience: Computer Vision, Speech Recognition, Predictive analysis, NLP

- **Person re-identification and tracking using multiple CCTV camera feeds.** 📄

- Developed a system to re-identify, track, and monitor customers' movement in Retail outlets from CCTV camera.
- Person Detection using YOLOv5 model trained on a custom dataset, and Re-identification using embedding generated by a Resnet-50 model trained on triplet loss.
- Developed a 2D projection algorithm to map each persons' position from camera image to floorplan.

- **Facial Recognition based automatic attendance system.** 📄

- Implemented face detection using Retinaface and face recognition using Arcface.
- Converted the model to Tflite to improve inference speed by 4×.
- Designed the database (MySQL) for daily attendance entry and built a dashboard with Django.

- **Medicine name recognition system from handwritten prescription.** 📄

- Developed a ROI Extraction model by using LayoutLM.
- Developed OCR models using EasyOCR, PyTesseract, and PaddleOCR to recognize medicine names.

- **Voice Controlled Conversational Chatbot.** 📄

- Implemented a streaming ASR model and a Text-to-speech model for automated speech recognition of users.

- Designed a sliding window technique to provide the ASR model with real-time streaming capabilities.

Research and Development Executive

Dhaka, Bangladesh

- Spectrum Engineering Consortium Limited (Full-time)

April 2021 - Sep 2021

Domain Experience : Robotics, Computer Vision, Microcontroller, IoT

- Developed a prototype of a low-cost mobile device, that can assist a visually disabled person to read, recognize person, and detect objects using voice command.
- Worked with various Microcontroller devices (Arduino, Raspberry-pi, ATMEGA, Nvidia Jetson Nano), Wireless communication modules (NRF24L01, HC-05), Hybrid Stepper Motor and drivers.

EDUCATION

The Ohio State University

Columbus, Ohio, USA

- Ph.D. in Computer Science and Engineering (CSE)

August 2023 - 2028 (Expected)

Selected Courses: Artificial Intelligence, Speech and Language Processing, Neural Networks

Bangladesh University of Engineering and Technology (BUET)

Dhaka, Bangladesh

- BSc. in Electrical and Electronic Engineering (EEE)

Feb 2016 - Feb 2021

Selected Courses: Digital Signal processing, Microcontroller, Wireless communication

PUBLICATIONS

Published

- Charig Yang, Samiul Alam, **Shakhrul Iman Siam**, Michael J. Proulx, Lambert Mathias, Kiran Somasundaram, James Fort, Omkar Parkhi, Yuheng Ren, Mi Zhang, Yuning Chai, Richard Newcombe, Hyo Jin Kim. "Reading Recognition in the Wild" (NeurIPS 2025). [Project Link]
- **Shakhrul Iman Siam**, Hyunho Ahn, Li Liu, Samiul Alam, Hui Shen, Zhichao Cao, Ness Shroff, Bhaskar Krishnamachari, Mani Srivastava, and Mi Zhang. "Artificial Intelligence of Things: A Survey" ACM Transactions on Sensor Networks (TOSN 2024). Paper Link
- Chenning Li, Yidong Ren, Shuai Tong, **Shakhrul Iman Siam**, Mi Zhang, Jiliang Wang, Yunhao Liu, and Zhichao Cao. "ChirpTransformer: Versatile LoRa Encoding for Low-power Wide-area IoT." In Proceedings of the 22nd Annual International Conference on Mobile Systems, Applications and Services (Mobisys 2024). [Paper Link]
- **Md Shakhrul Iman Siam**, and Subrata Biswas. "A Deep Learning Based Person Detection and Heatmap Generation Technique with a Multi-Camera System." In 2022 12th International Conference on Electrical and Computer Engineering (ICECE 2022).
- **Md Shakhrul Iman Siam**, Md Saiful Bari Siddiqui, Mushfiqul Abedin, and Mohammed Imamul Hassan Bhuiyan. "Bioradiolocation-Based Multi-Class Sleep Stage Classification Using Time and Frequency Features with Random Forest Classifier." In 2022 12th International Conference on Electrical and Computer Engineering (ICECE 2022).
- KM Naimul Hassan, Subrata Kumar Biswas, Md Shakil Anwar, **Md Shakhrul Iman Siam**, and Celia Shahnaz. "A Dual-Purpose Refreshable Braille Display Based on Real Time Object Detection and Optical Character Recognition." In 2019 IEEE International Conference on Signal Processing, Information, Communication & Systems (SPICSCON 2019).

Under Review

- **Shakhrul Iman Siam**, Tiantian Feng, Jiankun Zhang, Shrikanth Narayanan and Mi Zhang. "RespiraMFM: A Multimodal Foundation Model with Contrastive Audio-Language Alignment for Respiratory Disease Identification." (NAACL 2026).
- **Shakhrul Iman Siam**, Dushyant Sharma, Stanislav Yu. Kruchinin, and Peter Skala. "DECAF: Disentangling Encoder for Compressed Audio Features" (ICASSP 2026).
- Hyunho Ahn, Hansol Lee, **Shakhrul Iman Siam**, Sachin Kumar. "How Much Context Is Enough? Evaluating the Role of Audio and Textual Context in ASR Systems." (ICASSP 2026).

SKILLS SUMMARY

Programming Languages	Python, C, C++, JavaScript, MATLAB, Assembly(ARM), Unix, Shell Scripts
Machine Learning Domain	Computer Vision, Generative AI, Foundation Model, Face-Recognition, Predictive Analysis, Recommendation System, Chatbot, Reinforcement Learning
ML-DL Frameworks	Pytorch, TensorFlow, Keras, Scikit-learn, NLTK, Pandas, OpenCV, Numpy, Unity
LLM Frameworks	MLC-LLM, DeepSpeed
Backend Frameworks	Django, Django Rest Framework
Databases	MySQL, Oracle
Web Development	HTML, CSS, Bootstrap
Cloud & Containerization	AWS, Docker, Kubernetes
Hardware & IoT	Arduino, Raspberry pi, Microcontroller, Meta Aria Glasses

SELECTED PROJECTS

- **Generating Customer Heatmap of a retail outlet from CCTV camera :** Using Yolov5 object detection model, Homography transformation and Kernel density estimator to generate customer density map on a floorplan of a retail outlet using video taken from CCTV camera. [↗](#)
- **Blind vision Assistance :** Prototype of a low-cost mobile device, that can assist a visually disabled person to read, recognize person, and detect objects using voice command. There are three main features of this system- Object detection, Face Recognition, and optical character recognition. [↗](#)
- **Real-time vehicle detection and tracking at junction using a fisheye camera :** The Video and Image Processing cup 2020 challenge focuses on fisheye cameras mounted into street lamps at junctions and vehicle detection and tracking to be used for a junction management system to optimize the flow of traffic and synchronize with other junctions to obtain bottleneck performances throughout the city. [↗](#)
- **Open Source Contribution in developing a pypi package:** A Python library for offline reverse geocoding. Reverse Geocode BD takes a (latitude , longitude) coordinate and returns the Division, District, Upazila and Thana of any location in Bangladesh. [↗](#)
- **Unsupervised Synthetic/Fake Speech Detection.** Our research introduces a novel unsupervised model, which we have termed "Anocoder". This model employs a one-dimensional convolutional autoencoder and is designed to differentiate between authentic and synthetic or fabricated speech.
- **Web Scrapping and Sentiment Analysis from Social Media post:** Using Selenium for web scrapping and analysis of social media and online newspaper posts for sentiment analysis. Tech: Selenium, BeautifulSoup

HONORS AND AWARDS

- First Runner up at Robi **Datathon 2.0** (2022)
- 5th at **IEEE Video and Image Processing Cup** (2020)
- Champion of Bangladesh Section and World Finalist, **IEEE YESIST12 Innovation Challenge** (2019)
- 1st Runner up, BUET CSE Day **Math Olympiad** (2017)
- 1st Runner up, BAS Divisional **Science Olympiad** (2015)
- Champion , Dutch Bangla Bank Prothom Alo Regional **Math Olympiad** (2015)
- 2nd Runner Up , Dutch Bangla Bank Prothom Alo Regional **Math Olympiad** (2014)
- Champion , Dutch Bangla Bank Prothom Alo Regional **Math Olympiad** (2011)

REFERENCE

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